Stochastic Thinking & Monte Carlo Simulation

(download slides and .py files to follow along!)

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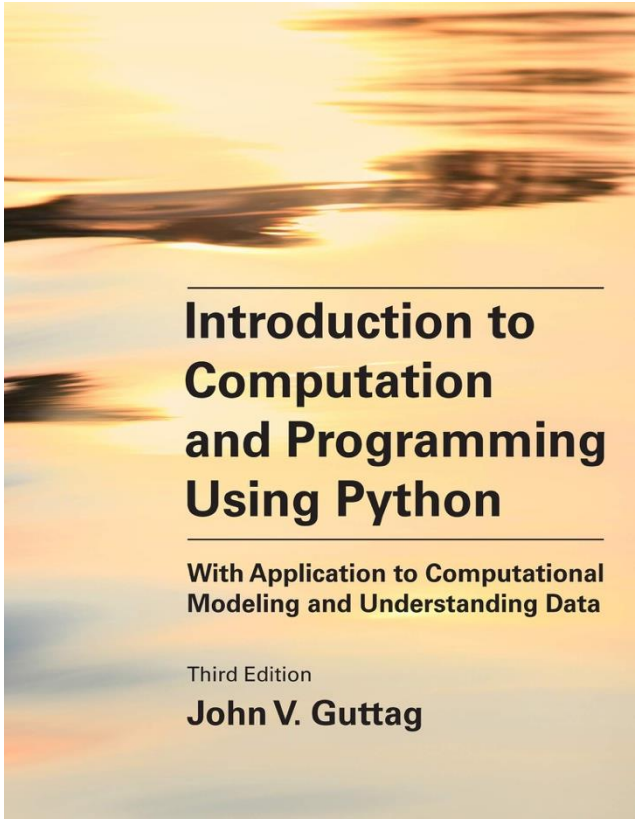
Happy Lunar New Year

Assigned Reading

- Today:
  - Sections 17.1 – 17.3
- Next lecture:
  - Chapter 16



All code and an errata sheet at
https://github.com/guttag/Intro-to-Computation-and-Programming

Asking Questions During Lecture

- Seems to be some reluctance to raise your hand and ask a question
- Perhaps easier to send an anonymous message

Lectures

Mon and Wed
11-12:30pm in 26-100
Materials in calendar below
Slido for questions during lecture

- TA's will monitor and ask questions on your behalf

Today's Lecture

- Randomness and simulation

SMART NEWS

Mathematician Who Made Sense of the Universe's Randomness Wins Math's Top Prize

Michel Talagrand took home the 2024 Abel Prize for his work on stochastic systems, randomness and a proof of a physics reaction that many experts thought was unsolvable

Top prizes in both Math and CS Awarded in 2024

Computer Theorist Wins $1 Million Turing Award

This year's honor will go to Avi Wigderson, an Israeli-born mathematician and theoretical computer scientist who specializes in randomness.

The Simple World of Newtonian Mechanics

- For thousands of years, people have attempted to build models to explain the physical world that they could observe with their eyes
- The core belief was that every effect has a cause, so physical world can be understood causally
  - E.g., Newton's three laws of motion
  - Suggests that relatively simple mathematical descriptions of physical effects are possible

1643 – 1727

While We are on the Subject of Issac Newton

- Plague forced closure of Cambridge University campus from 1665– 1667
- Newton decamped to his family's country house
- While there
  - He invented calculus (18.01, 18.02)
  - Derived first version of laws of gravity (8.01)
  - Formulated foundations of optics of color

Which Raises the Questions

- What did you accomplish when stuck at home during Covid?
- What did I accomplish when MIT campus was closed from April 2020 to August?

I hate to brag,

but I got dressed every day (almost).

Two+ Centuries After Newton

- In early 20th century, atomic and subatomic observations seemed to defy classical mechanical explanations
  - To explain the photoelectric effect, Einstein postulated that light acts both as a particle (photon) and as a wave
- Led to introduction of quantum mechanics, and assertion that one cannot precisely measure a particle's physical properties

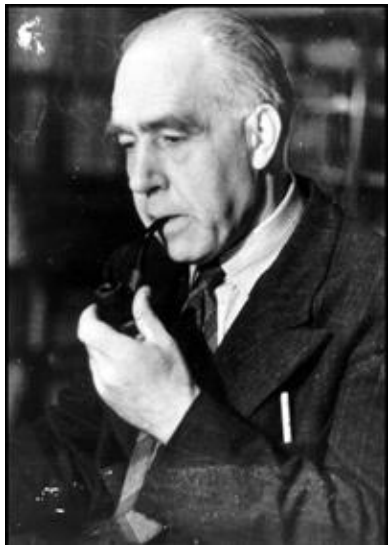
- 20th century physics
  - 8.04, 8.05, 8.06

Two+ Centuries After Newton

- Heisenberg and Bohr argued that at its most fundamental level, the behavior of the physical world cannot be predicted
  - Heisenberg uncertainty principle
- Fine to make statements of the form "x is highly likely to occur," but not of the form "x is certain to occur"
- Introducing uncertainty into modeling physical world

"The more precise the measurement of position, the more imprecise the measurement of momentum, and vice versa."

"Those who are not shocked when they first come across quantum theory cannot possibly have understood it. …If you think you can talk about quantum theory without feeling dizzy, you haven't understood the first thing about it."



- 8.04, 8.05

Not Everyone Bought It

- The philosophical implications of being unable measure things precisely or understand the specific cause of an observed effect were unsettling to some
- Among others, Einstein and Schrodinger objected
  - "I, at any rate, am convinced that He [God] does not throw dice."– Albert Einstein
  - "Einstein, don't tell God what to do."– Niels Bohr

"The demand for certainty is one which is natural to man, but is nevertheless and intellectual vice"
– Bertrand Russell

Does It Really Matter?

- Suppose I flip two coins
- Can you correctly predict whether the flips will yield
  - 2 heads?
  - 2 tails?
  - 1 head and 1 tail?
- Need to know accurately:
  - Weight distribution of coin
  - Velocity and acceleration of thumb
  - Orientation of coin on thumb before flip
  - Air flow around coin
  - Height above landing spot
  - Elasticity of floor
  - …

The Moral

- The world might or might not be inherently unpredictable
- Even if not, lack of knowledge prevents precise predictions
- For example, suppose I have already flipped coins?
- Therefore, can treat the world as inherently unpredictable

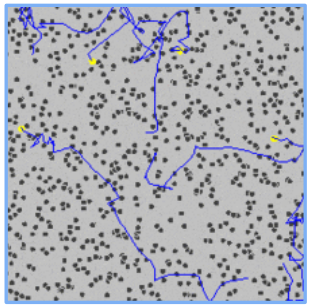
Causal nondeterminism – some events truly random

Predictive nondeterminism – in principle might be able to predict, but don't have enough information. There is chaos, but not randomness

Use a **stochastic** process to represent systems or to model phenomena that seem to change in a random way

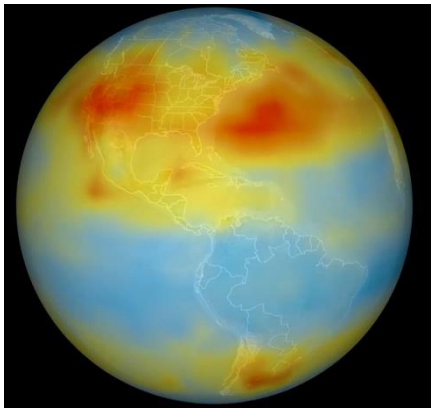
Some Places We Use Stochastic Models

- Systems with choices in state transitions, e.g.,
  - Motion of particles in a fluid (Brownian motion)
  - Outcomes of play in football



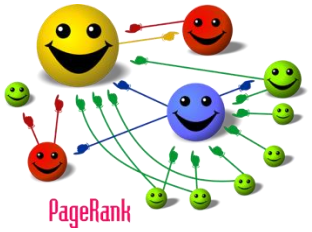
- Systems where measurement is uncertain or noisy, e.g.,
  - Polling data
  - Radio astronomy

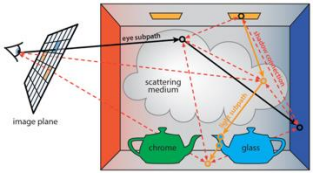
- Systems where can't measure all factors, e.g.,
  - Weather systems

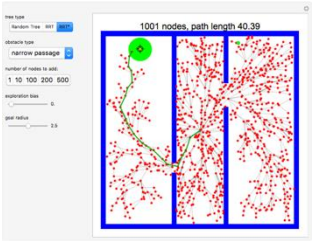


Stochasticity and Randomness in Computing

- Stochastic Gradient Descent is at the heart of deep learning and modern AI
- Markov Chain Monte Carlo methods are used throughout inference
- Simulated annealing uses randomness to find optimal solutions
- Large language models
- 3D rendering uses Monte-Carlo random sampling
- Simulation of complex systems
- Perturbation analysis uses randomness to characterize the sensitivity of systems and methods
- Randomized algorithms

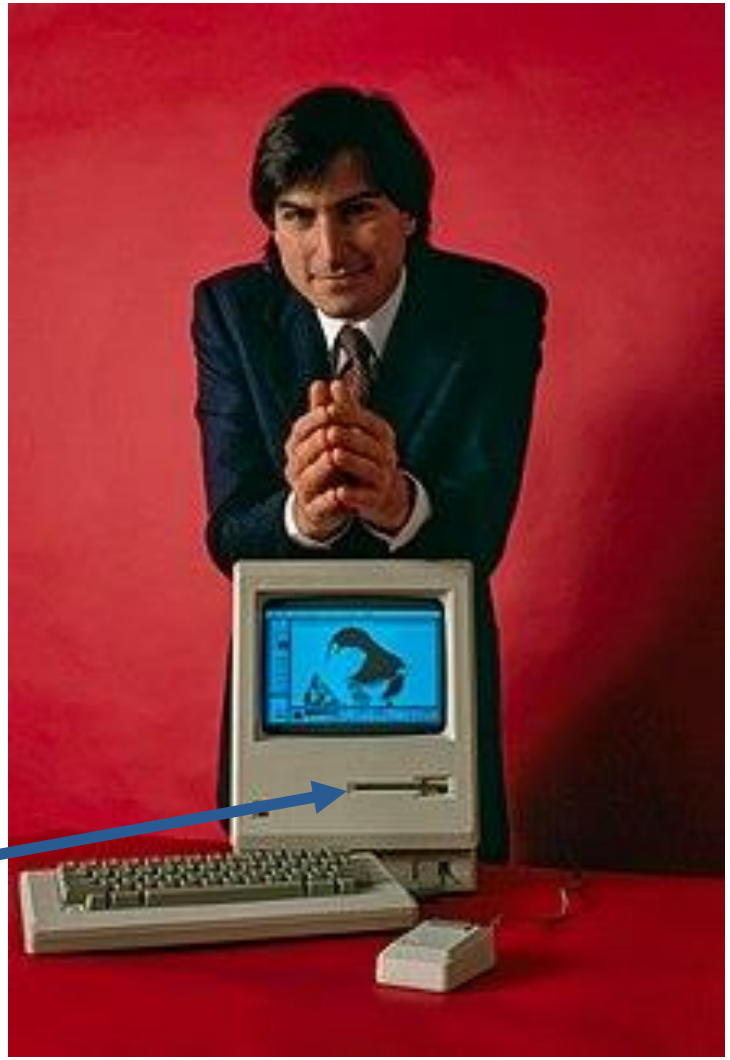






To Summarize

- If 6.100A led you to think that computer science and computing is mostly about deterministic processes,
- We're truly sorry
- Deterministic computing is *SO 20th century*



Steve Jobs and Macintosh Computer 1984

Stochastic Processes

- Systems often defined by a set of state variables, and processes that determine the transition to a next set of values for variables. It's a causal process.
  - E.g., a car's state at a point in time is described by a position, heading and velocity; its state at a subsequent time is determined by those parameters and actions of the driver
- In a **stochastic** process, the next state might depend on previous states **and on some seemingly random element**
  - E.g., a car's future position may not be exactly determined by its kinematics and the actions of the driver because of road conditions, topography, etc.
- This will require a change in how we think about building computational models of the world

Two Specifications for a Process

```
def roll_die():
    """Return an int between 1 and 6"""
    return 2
```

`random.choice` selects an element from a list with equal likelihood

```
def roll_die():
    """Return a random int between 1 and 6"""
    return random.choice([1, 2, 3, 4, 5, 6])
```

Any implementation that satisfies the second specification would also satisfy the first.

But one that satisfies the first specification might or might not satisfy the second

Simulate Five Rolls

```
def roll_die():
    """Return a random int between 1 and 6"""
    return random.choice([1, 2, 3, 4, 5, 6])

def test_roll(n):
    result = ''
    for _ in range(n):
        result += ' ' + str(roll_die())
    return result

for _ in range(10):
    print(test_roll(5))
```

How probable is the output 11111?

Discrete Probability is About Counting

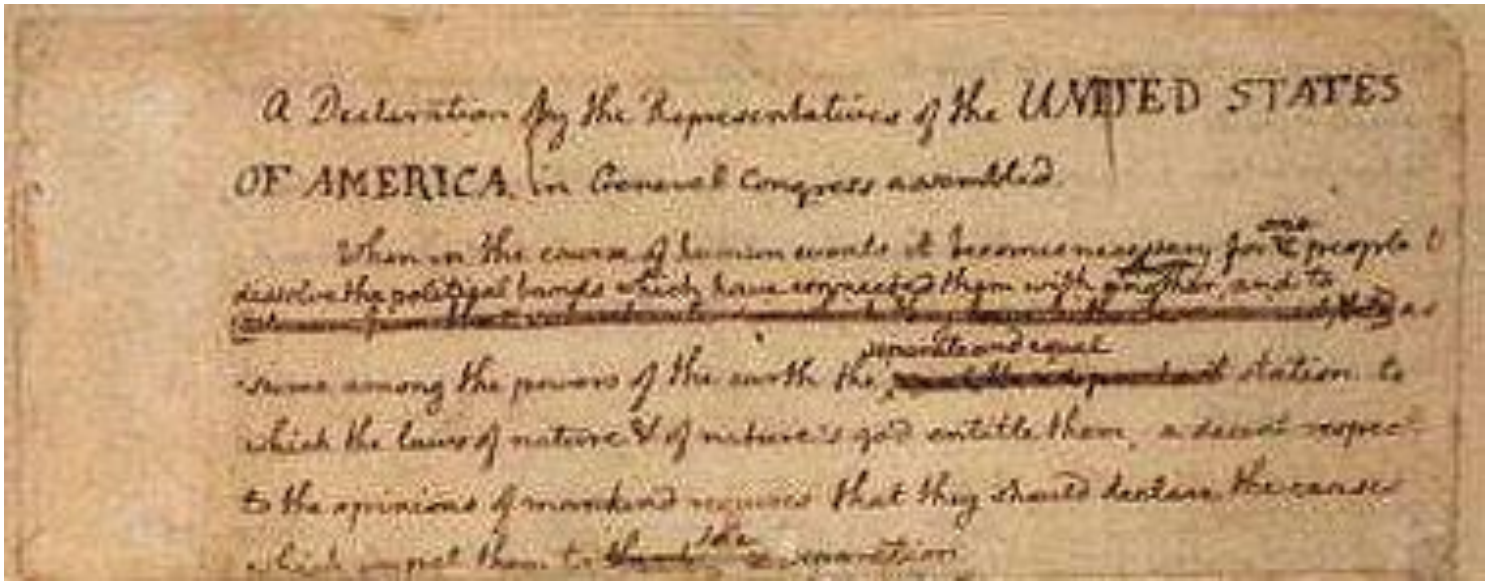
- Count the total number of possible events (often called the universe of events)
- Count the number of events that have the property of interest
- Divide second number by the first
- Probability of rolling 11111?
  - All events: 11111, 11112, 11113, …, 11121, 11122, …, 66666
  - Ratio: 1/(6**5)
  - ~0.0001286
- Each combination of five digits has the same probability

Some Basic Facts about Probability

- Probabilities are always in the range **0 to 1**.
0 if impossible, and 1 if guaranteed
- If the probability of an event occurring is **p**, the probability of it not occurring must be **1-p**
- When estimating probabilities that involve multiple events, must start with the question of whether the events are **independent** of each other
- More about computing probabilities next week

Independence

- Two events are **independent** if the outcome of one event has no influence on the outcome of the other
- Independence should not be taken for granted
  - Is your score on the next micro quiz independent of whether you pay attention in lecture?


A Declaration by the Representatives of the UNITED STATES OF AMERICA, in General Congress assembled.

When in the course of human events it becomes necessary for the people to dissolve the political bands which have connected them with another, and to assume among the powers of the earth the separate and equal station to which the laws of nature & of nature's god entitle them, a decent respect to the opinions of mankind requires that they should declare the causes which impel them to the separation.

Using Simulation to Estimate Probabilities

Using Simulation to Estimate Probabilities

- Going to explore simulation of rolling a group of dice many times
- First of many simulations we will see
- **Simulation**
  - Run many trials in which we select one event from the universe of possible events
  - For each trial, compute some properties of event
  - Report some statistics about the properties over the set of trials

A Simulation of Die Rolling

Simulate trying to roll a specific sequence

```
def run_sim(goal, num_trials):
    total = 0
    for i in range(num_trials):
        if i != 0 and i % 100_000 == 0:
            print(f"Starting trial {i}")
        result = ''
        for _ in range(len(goal)):
            result += str(roll_die())
        if result == goal:
            total += 1
    print(f"Actual probability of {goal} = "
          f"{1 / 6**len(goal):.8f}")
    print(f"Estimated Probability of {goal} = "
          f"{total / num_trials:.8f}")

run_sim('11111', 1000)
```

Repeat for many trials

Roll appropriate # of dice

Count # of hits

Compute observed probability

Output of Simulation

Actual probability of 11111 = 0.00012860
Estimated Probability of 11111 = 0.00000000

- Why did simulation give the **wrong** answer?
  - Rounding?
  - Coding bug?
  - Something else?

Why Did Simulation Give Wrong Answer?

- 6**5 = 7776 possibilities, so in 1000 trials we are unlikely to observe one event in that universe with enough frequency to estimate accurately
- Even if observed once, estimate would then be 0.001. I.e., still wrong.

Let's try 1,000,000 trials

Actual probability of 11111 = 0.0001286
Estimated Probability of 11111 = 0.000131

How about 10M trials

Actual probability of 11111 = 0.0001286
Estimated Probability of 11111 = 0.0001286

Morals

Aesop
620 BC –
564 BC

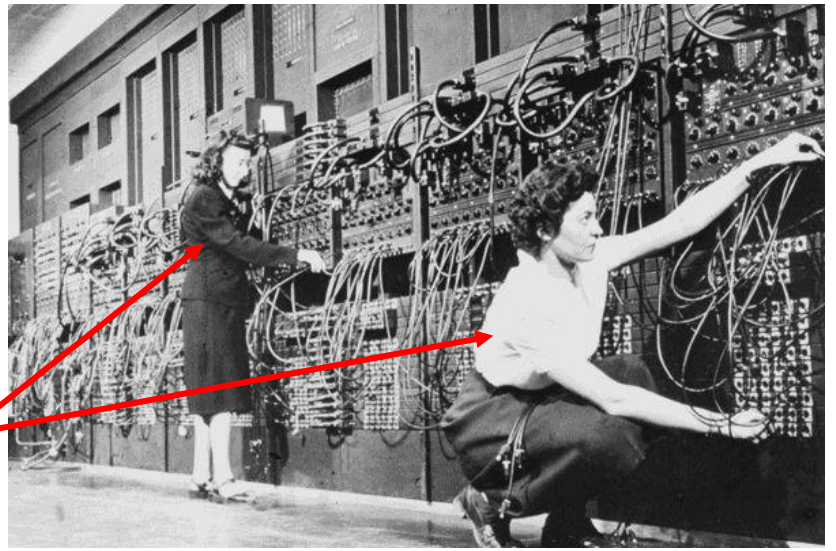
- Moral 1: it takes a lot of trials to get a good estimate of the frequency of occurrence of a **rare** event. We'll talk lots more later about how to **know** when we have enough trials to trust the estimate
- Moral 2: one should not confuse the **sample probability** with the actual probability
- Moral 3: there was really no need to do this by simulation, since there is a perfectly good closed form answer. We will see many examples where this is **not** true, where simulation is best way to provide answers about processes and outcomes

Simulations Are Used a Lot

- To model systems that are mathematically intractable
  - Otherwise may not be possible to analyze a system
- To extract useful intermediate results
- To support iterative development by successive refinement
  - Start with basic system, then incrementally add features
- To support exploring of variations by asking/answering "what if" questions

Monte Carlo Simulations

- Stanislaw Ulam, recovering from an illness, was playing a lot of solitaire (1946)
- Tried to figure out probability of winning, and failed
- Thought about playing lots of hands and counting number of wins (i.e., sampling)
  - ~10,000 hands needed
- Asked Von Neumann if he could build a program to simulate many hands on ENIAC
  - Creation of Monte Carlo simulation method



Programming, circa 1946

Monte Carlo Simulation

- A method of estimating the value of an unknown quantity using the principles of inferential statistics
- Inferential statistics
  - *Population*: a set of examples
  - *Sample*: a proper **subset** of a population
  - Key fact: a *random sample* tends to exhibit the same properties as the population from which it is drawn
    - Provided size of sample is sufficiently large and random
- The key question: when should we believe that a sample has the same properties as the population?

It's All About Confidence

- Consider flipping a coin to decide whether it is fair or not
- **Confidence** in our estimate depends upon two things
- **Size** of sample (e.g., 100 versus 2)
- **Variance** of sample (e.g., all heads versus 52 heads)
- As the **variance** grows, we need **larger** (or more) samples to have the same degree of **confidence**

A Monte Carlo simulation

- Want to explore relationship between **number of samples** and **confidence** in estimate of parameter
- In the past, I've used **roulette** as an example
  - Result of each spin is random
  - Since we know what actual estimates should be, gives us a way to explore how **confidence** in an estimate of some parameter (e.g., expected winnings) changes as the **number of samples** changes
- But this raises a moral dilemma

How Many Vices Should One 6 Unit Subject Cover?

- I already spent multiple lectures talking about how to be a successful burglar
- Do I really want to move onto gambling?

Roulette as An Example

No need to simulate, since answer is calculable

Allows us to compare simulation results to actual probabilities

If you are not familiar with roulette:

- Place a bet on a number (between 1 and 36)
- Croupier spins wheel then throws ball in opposite direction of spin
- When ball comes to rest in a pocket, bets on that number pay off
- Can also bet on color of pocket, other more complicated bets
- We will focus just on pocket number for this lecture

A Useful Class

```
class Fair_roulette():

    def __init__(self):
        self.pockets = list(range(1, 37))
        self.ball = None
        self.pocket_odds = len(self.pockets) - 1

    def spin(self):
        self.ball = random.choice(self.pockets)

    def bet_pocket(self, pocket, amt):
        if str(pocket) == str(self.ball):
            return amt * self.pocket_odds
        else:
            return -amt

    def __str__(self):
        return "Fair roulette"
```

Pockets range from 1 to 36

Equally likely to hit any pocket

If bet right pocket, then win

Amount won much larger than amount lost; but balanced by relative odds

Monte Carlo Simulation

Allow for different versions of roulette

```
def play_roulette(game, num_spins, pocket, bet):
    tot_pocket = 0
    for _ in range(num_spins):
        game.spin()
        tot_pocket += game.bet_pocket(pocket, bet)
    return tot_pocket
```

Keep track of overall winnings

Test Code

```
def format_dollar(amount):
    return f"{'-' if amount < 0 else ''}${abs(amount):.2f}"

def test_roulette(game, num_spins, num_trials):
    game_name = type(game).__name__
    bet = 10
    winnings = 0
    for _ in range(num_trials):
        residual = play_roulette(game, num_spins, 2, bet)
        winnings += residual
        if num_trials < 11:
            print(f'Return on betting ${bet} on a pocket for {game_name} '
                  f'{num_spins} times = {format_dollar(residual)}\n')
    print(f'Average return = {format_dollar(winnings)}')

random.seed(0)
test_roulette(Fair_roulette(), 10, 3)
```

Three Trials of 10 Spins of the Wheel

```
test_roulette(Fair_roulette(), 10, 3)
```

```
Return on betting $10 on a pocket for Fair_roulette 10
times = -$100.00

Return on betting $10 on a pocket for Fair_roulette 10
times = -$100.00

Return on betting $10 on a pocket for Fair_roulette 10
times = $260.00

Average return = $60.00
```

- Should we trust this average?
- High variation across trials

100 trials of 10 Spins of the Wheel

```
Average return = -$640.00
```

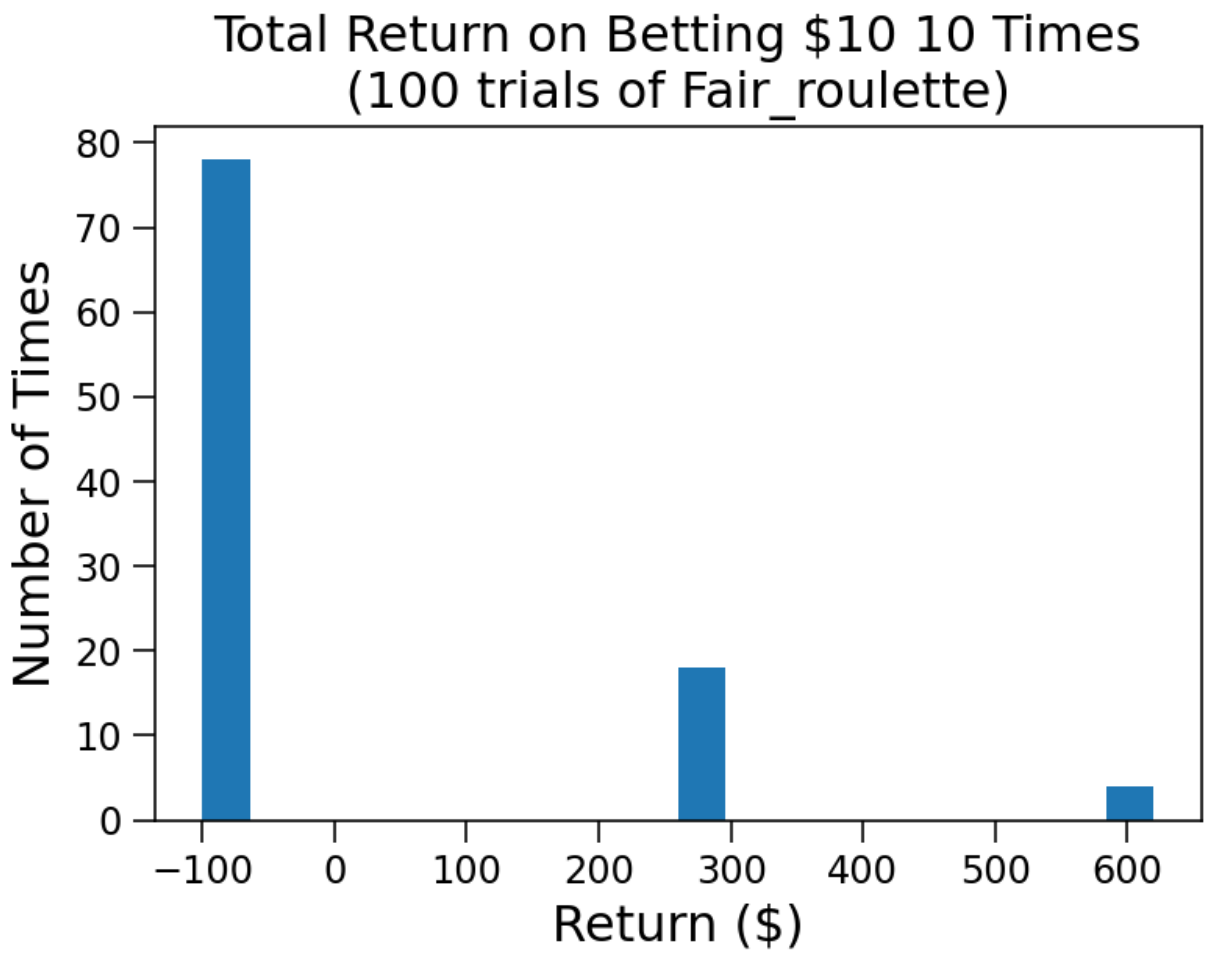
- Should we trust this average?
- What does Variation look like across 100 trials?

Looking at the Distribution of Possible Outcomes

```
def test_roulette(game, num_spins, num_trials):
    game_name = type(game).__name__
    bet = 10
    winnings = []
    for _ in range(num_trials):
        winnings.append(play_roulette(game, num_spins, 2, bet))
    amt = sum(winnings)/(len(winnings)*num_spins)
    ave_return = f"{'-' if amt < 0 else ''}${abs(amt):.2f}"
    print(f'Average return per ${bet} bet over {num_spins} '
          f'spins of {game_name} = {ave_return}')
    plt.hist(winnings, bins = 20)
    plt.title(f'Total Return on Betting $10 {num_spins} Times\n'
              f'({num_trials} trials of {num_spins} '
              f'spins of {game_name})')
    plt.xlabel('Return ($)')
    plt.ylabel('Number of Times')
```

100 trials of 10 Spins of the Wheel

```
Ave. return per $10 bet over 10 spins of Fair_roulette = -$0.64
```

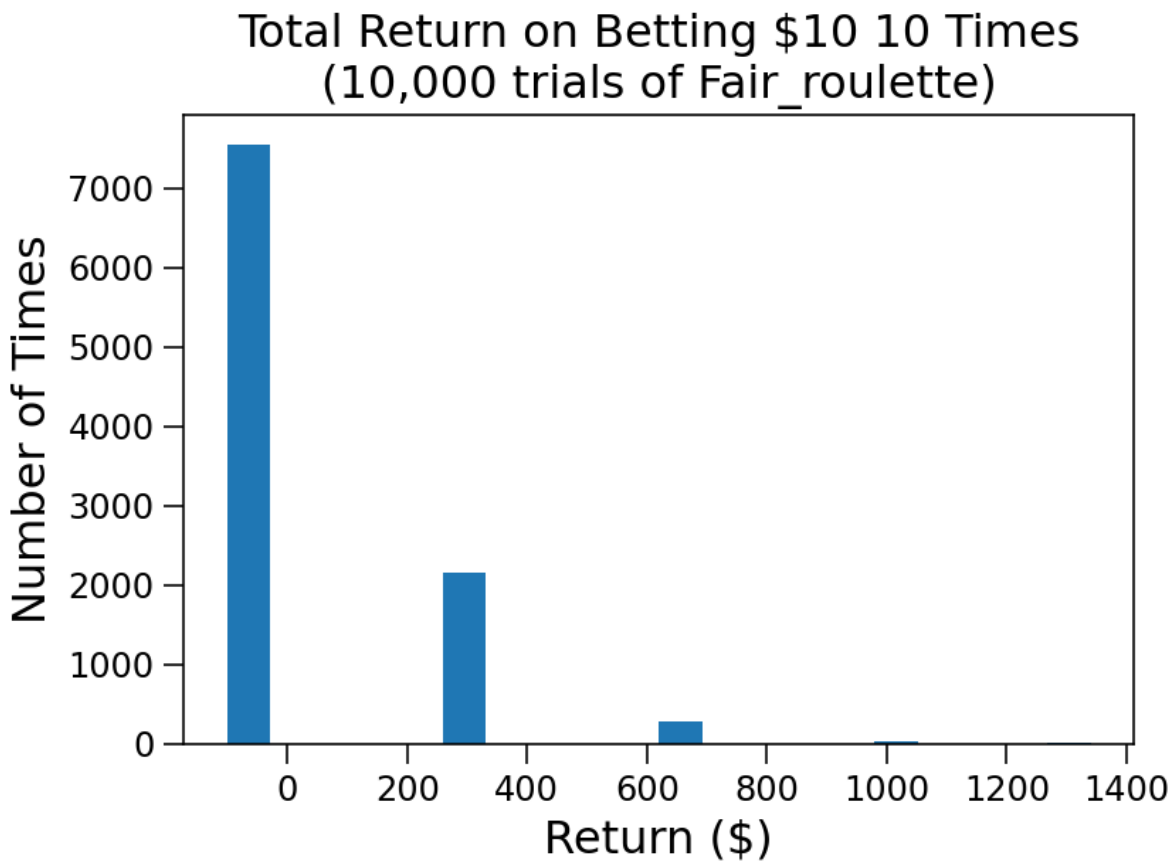


- Why only 3 bars?

0.00002143

10,000 Trials of 10 Spins of the Wheel

```
Ave. return per $10 bet over 10 spins of Fair_roulette = $0.05
```

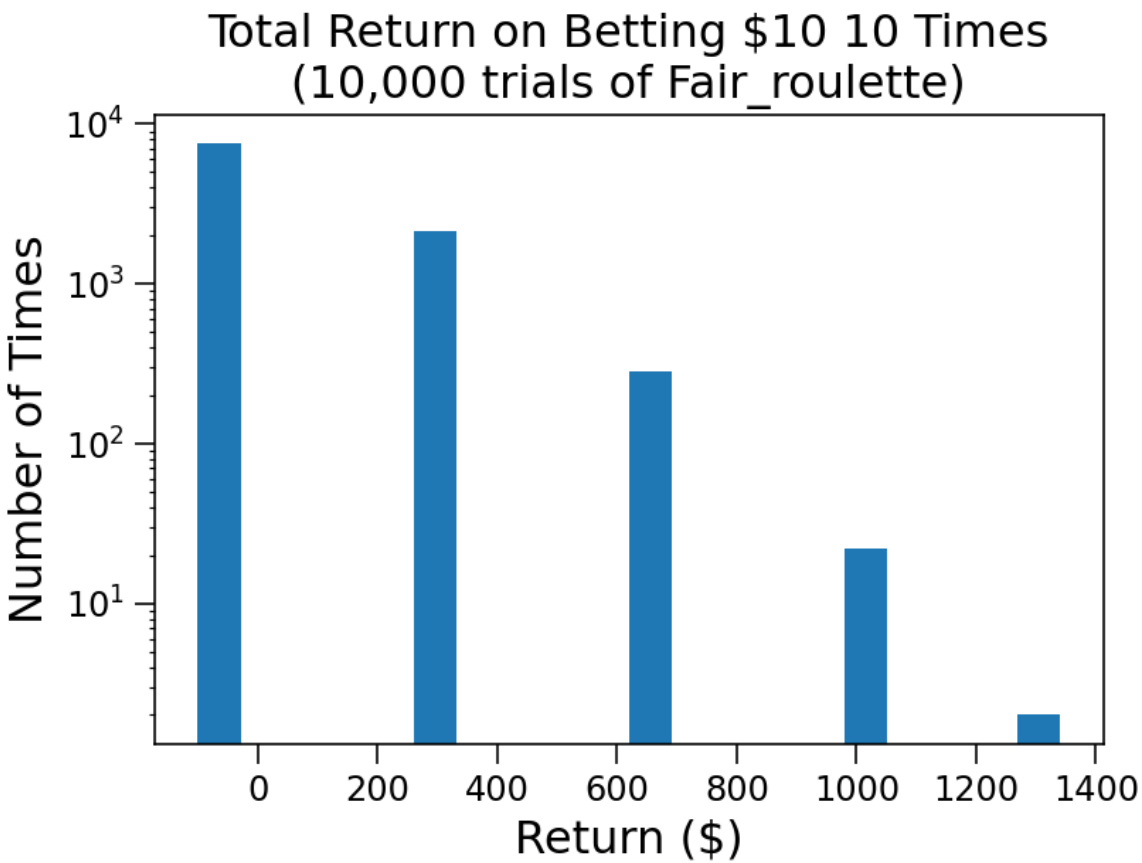


How many bars?

Try semilog y

10,000 Trials of 10 Spins of the Wheel

```
Ave. return per $10 bet over 10 spins of Fair_roulette = $0.05
```



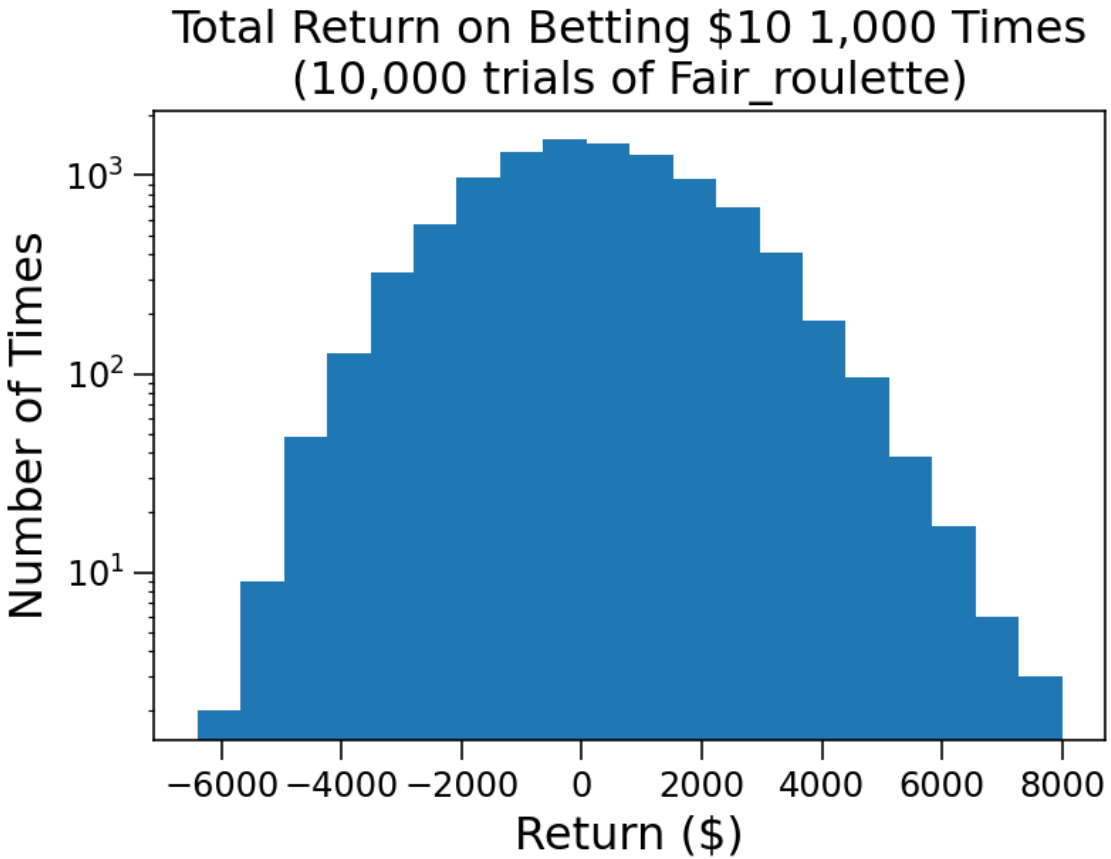
On average, close to break even

But that doesn't come to telling the whole story

Most common outcome: go home broke

10,000 Trials of 1000 Spins of the Wheel

```
Ave. return per $10 bet over 1000 spins in Fair_roulette = $0.02
```



On average, even closer to break even

Near break even more often than not

Still good days and bad days

The Take Away

- As we increase number of samples per trial, we get a better estimate of mean
- As we increase number of trials, we get a better sense of distribution of possible results
- Both are important!
- And they are different!

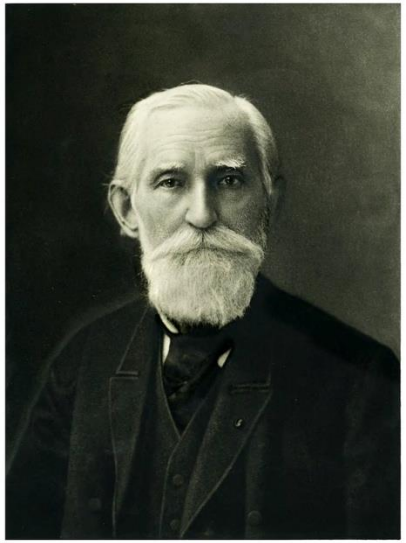
Why the Sample Mean Converges

- The average of a set of samples drawn with the same given probability p from a population converges to the expected value of the population when the number of samples goes to infinity

Jacob Bernoulli

Borel + Chebyshev Law of Large Numbers

- In repeated independent tests with the same actual probability *p* of a particular outcome in each test, the chance that the fraction of times that outcome occurs differs from *p* converges to zero as the sample size goes to infinity



Does this imply that if deviations from expected behavior occur, these deviations are likely to be ***evened out*** by **opposite** deviations in the future?

Gambler's Fallacy

- "On August 18, 1913, at the casino in Monte Carlo, black came up a record twenty-six times in succession [in roulette]. … [There] was a near-panicky rush to bet on red, beginning about the time black had come up a phenomenal fifteen times." -- Huff and Geis, *How to Take a Chance*

- Probability of 26 consecutive black
  1/67,108,865 = 1.49e-8
- Probability of 26 consecutive black when previous 25 rolls were black
  1/2
- In **independent** trials, past results have no effect on future trials

Break

Related but Different: Regression to the Mean

TABLE I.

NUMBER OF ADULT CHILDREN OF VARIOUS STATURES BORN OF 205 MID-PARENTS OF VARIOUS STATURES.
(All Female heights have been multiplied by 1·08).

Heights of the Mid-parents in inches.	Heights of the Adult Children.														Total Number of		Medians.
	Below	62·2	63·2	64·2	65·2	66·2	67·2	68·2	69·2	70·2	71·2	72·2	73·2	Above	Adult Children.	Mid-parents.	
Above ..	..	..	..	..	..	..	..	..	..	..	..	1	3	..	4	5	..
72·5	..	..	..	..	..	..	..	1	2	1	2	7	2	4	19	6	72·2
71·5	..	..	..	..	1	3	4	3	5	10	4	9	2	2	43	11	69·9
70·5	1	..	1	..	1	1	3	12	18	14	7	4	3	3	68	22	69·5
69·5	..	..	1	16	4	17	27	20	33	25	20	11	4	5	183	41	68·9
68·5	1	..	7	11	16	25	31	34	48	21	18	4	3	..	219	49	68·2
67·5	..	3	5	14	15	36	38	28	38	19	11	4	..	..	211	33	67·6
66·5	..	3	3	5	2	17	17	14	13	4	..	..	..	..	78	20	67·2
65·5	1	..	9	5	7	11	11	7	7	5	2	1	..	..	66	12	66·7
64·5	1	1	4	4	1	5	5	..	2	..	..	..	..	..	23	5	65·8
Below ..	1	..	2	4	1	2	2	1	1	..	..	..	..	..	14	1	..
Totals ..	5	7	32	59	48	117	138	120	167	99	64	41	17	14	928	205	..
Medians ..	..	..	66·3	67·8	67·9	67·7	67·9	68·3	68·5	69·0	69·0	70·0	..	..	..	..	..

NOTE.—In calculating the Medians, the entries have been taken as referring to the middle of the squares in which they stand. The reason why the headings run 62·2, 63·2, &c., instead of 62·5, 63·5, &c., is that the observations are unequally distributed between 62 and 63, 63 and 64, &c., there being a strong bias in favour of integral inches. After careful consideration, I concluded that the headings, as adopted, best satisfied the conditions. This inequality was not apparent in the case of the Mid-parents.

Francis Galton, 1885

Data on height of adult children, versus average height of parents (where height of mother is adjusted to account for gender differences)

Regression to the Mean

- Galton observed that extreme characteristics (e.g., height) in parents are not passed on **completely** to their offspring
- Characteristics in the offspring *regress* towards a *"mediocre"* point (what we now call a mean)
- Through experiments, Galton quantified regression to the mean. He wrote that, *"the average regression of the offspring is a constant fraction of their respective mid-parental deviations."*
- Difference between a child and its parents for some characteristic is proportional to its parents' deviation from typical people in population.
  - If parents are each two inches taller than averages for men and women, on average child will be shorter than its parents by some factor (one minus the regression coefficient) times two inches
  - For height, Galton estimated the multiplier to be about 2/3

The James Family (LeBron, Savannah, Bronny)

```
L_James = 6*12 + 9
S_James = 5*12 + 7
Bronny = 6*12 + 4
ave_man = 5*12 + 9
ave_woman = 5*12 + 3.5
parent_ave = (S_James + L_James)/2
am_ave = (ave_woman + ave_man)/2
diff = parent_ave - am_ave
pred_Bronny = L_James - diff*(2/3)
print(f'Predicted height = {pred_Bronny:.2f} inches')
print(f'Actual height = {Bronny} inches')
```

Predicted height = 75.83 inches

Purported height = 76 inches

Closer than I expected

Regression to(ward) the Mean phenomenon

- Following an extreme random event, the next random event is likely to be less extreme
- If you spin a fair roulette wheel 10 times and get 100% reds, that is an extreme event (probability = 1/1024)
- It is likely that in the next 10 spins, you will get fewer than 10 reds
  - But the expected number is not less than 5 (!)
  - The expected number is 5
- So, if you look at the average of the 20 spins, it will be closer to the expected mean of 50% reds than to the 100% of the first 10 spins

Regression toward the Mean

- Following an extreme random event, the next **random** event is likely to be less extreme
- More interesting case: when events have random **and non-random components**
  - Consider the example of grades or athletics
  - Performance can be modeled as a combination of natural ability and some amount of luck

Simulate Students Taking a Quiz

```
def get_grade(student, num_questions):
    tot_right = 0
    for _ in range(num_questions):
        # student[i] is talent level. Higher is better
        if random.random() < student[1]:
            tot_right += 1
    return 10*(tot_right/num_questions)
```

name

talent

```
num_students = 200
students = [(f'student{i}', random.choice((0.3, 0.4, 0.5, 0.6, 0.7)))
            for i in range(num_students)]
grades = []
for s in students:
    grades.append((s, get_grade(s, 4)))
```

Best and Worst Students on Quiz 1

student139: 10	student120: 0
student140: 10	student122: 0
student141: 10	student137: 0
student173: 10	student154: 0
student176: 10	student22: 0
student178: 10	student23: 0
student195: 10	student3: 0
student197: 10	student44: 0
student80: 10	student46: 0
student97: 10	student59: 0

Mean for top 10 = 10.0, Mean for bottom 10 = 0.0

Big difference

Is it all due to talent?

Compare Same Groups on Quiz 2

Best on Q 1	Worst on Q 1
student139: 4	student120: 1
student140: 6	student122: 1
student141: 9	student137: 4
student173: 8	student154: 4
student176: 2	student22: 4
student178: 7	student23: 1
student195: 6	student3: 5
student197: 5	student44: 2
student80: 6	student46: 4
student97: 6	student59: 3

Still a difference

But not nearly as large

Mean on Q2 for top 10 on Q1 = 5.9,
Mean for bottom 10 on Q1 = 2.9

Regression Towards the Mean a Big Deal

- Regression towards the mean MUST be taken into account when designing experiments
- Much of the so-called “placebo effect” is just regression towards the mean
  - Select subjects for experiments based on their extreme condition (very sick, very high blood pressure, etc.)
  - Next measurement is likely to be better, even without intervention
- Hence the need for randomized control trials (RCT)

Regression to Mean Often the Better Explanation

- When one of his players had an unusually good game, the coach congratulated them
- When one of his players had an unusually bad game, he berated them
- Over time, he observed that criticism was typically followed by improvement, but praise was typically followed by worse performance
- Should he conclude that criticism is useful, but praise counter-productive?

Recap so far

- **Law of large numbers:** the more samples, the more things conform to the expected values and probabilities
- **Gamblers fallacy:** wrong idea that future events will have opposite behavior to even out probabilities
- **Regression towards the mean phenomenon:** after extreme measurements, the following observations are likely to be closer to average
  - Obvious for pure independent random events
  - Tends to be true in many partially independent and random events

Some Shocking News!

Casinos Not in the Business of Being Fair

European Wheel

American Wheel

Two Subclasses of Roulette

```
class Eu_roulette(Fair_roulette):

    def __init__(self):
        super().__init__()
        self.pockets.append('0')

    def __str__(self):
        return "European roulette"


class Am_roulette(Eu_roulette):

    def __init__(self):
        super().__init__()
        self.pockets.append('00')

    def __str__(self):
        return "American roulette"
```

Changed the number of pockets, but not the payoff on a winning bet

Comparing the Games Over 1,000 Trials

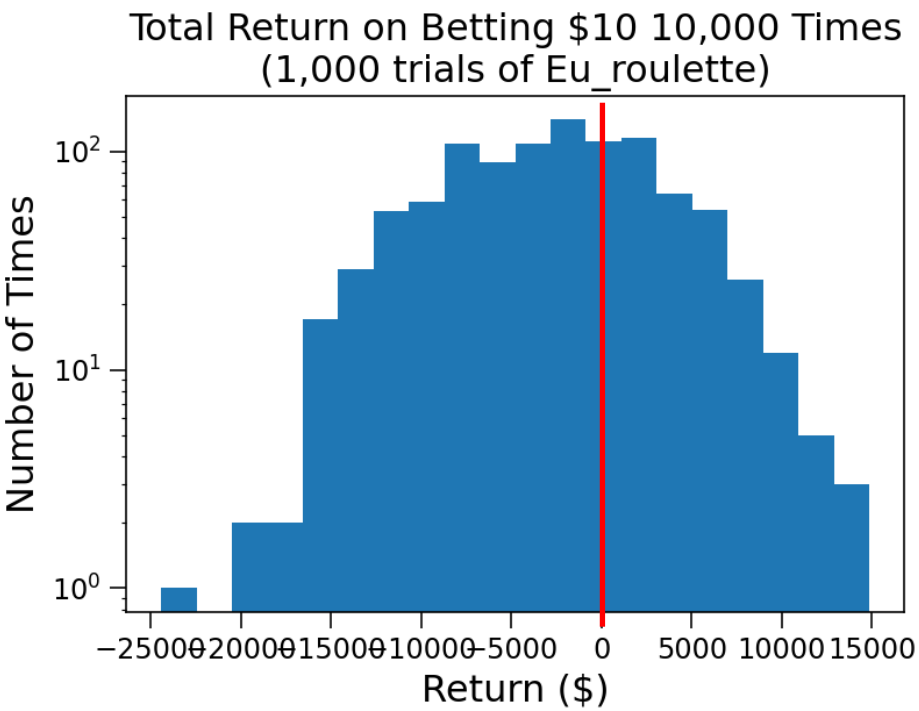
```
Ave. return per $10 bet over 10 spins of Fair_roulette = -$0.21
Ave. return per $10 bet over 10 spins of Eu_roulette = -$0.39
Ave. return per $10 bet over 10 spins of Am_roulette = -$0.86
Ave. return per $10 bet over 100 spins of Fair_roulette = $0.05
Ave. return per $10 bet over 100 spins of Eu_roulette = -$0.22
Ave. return per $10 bet over 100 spins of Am_roulette = -$0.61
Ave. return per $10 bet over 1,000 spins of Fair_roulette = $0.11
Ave. return per $10 bet over 1,000 spins of Eu_roulette = -$0.20
Ave. return per $10 bet over 1,000 spins of Am_roulette = -$0.47
Ave. return per $10 bet over 10,000 spins of Fair_roulette = $0.02
Ave. return per $10 bet over 10,000 spins of Eu_roulette = -$0.27
Ave. return per $10 bet over 10,000 spins of Am_roulette = -$0.51
```

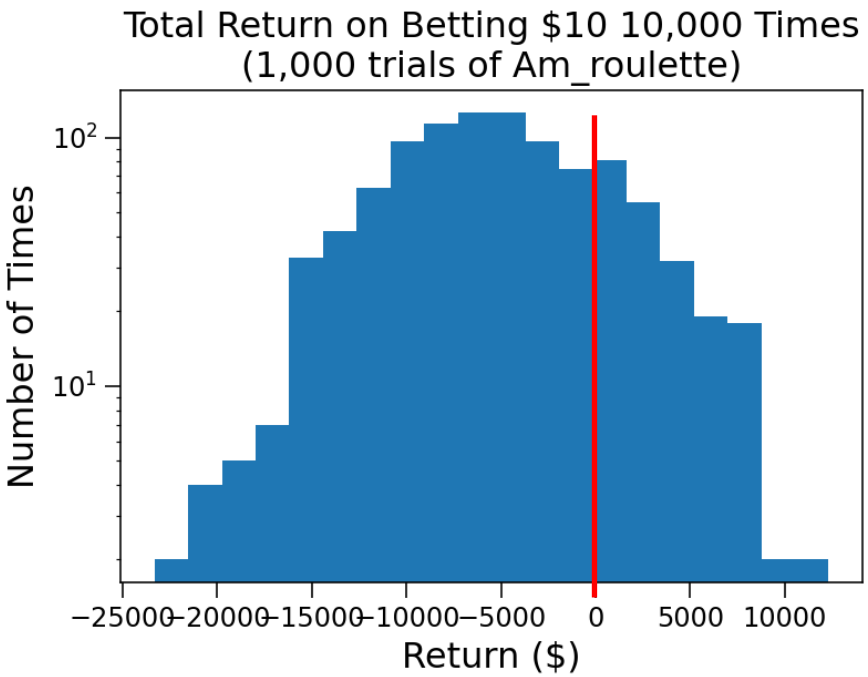
Can see that:

1) Eventually, the casino **always wins**!
2) As number of spins grows, the expected return seems to converge to match analytic expectation

Analytic Expectation
0.000
-.2703
-.5263

But It Is Possible to Win Sometimes





Does One Have to Lose When Playing Roulette?

Per a Popular Chatbot

The Martingale system is one of the oldest and most famous betting strategies for roulette.

How it works

If you **win**, you keep your profit and quit.

If you **lose**, you **double your bet** and try again.

When you eventually win, the win covers all past losses plus gives you a profit equal to your original bet.

I've written some code to try this out

Maybe It Works

Number of simulations = 1,000
For betting a pocket in American Roulette
Ave. spins before winning = 38.29
Worst case spins before winning = 256

If each spin took a minute,
On average, you're a winner in about 40 minutes
Worst case you're a winner in about 4.5 hours

Should you give it a try?

On the Other Hand

Ave. amount down before winning =
$578,962,655,284,616,254,631,692,635,733,651,768,663,180,971,491,815,733,635,303,740,654,441,463,808
Worst case amount lost before winning =
$578,960,446,186,580,977,117,854,925,043,439,539,266,349,923,328,202,820,197,287,920,039,565,648,199,670

You'd run out of money before you'd run out of time

Even if you were Elon Musk

Summary

- Don't play roulette
- Simulations highly useful for understanding stochastic situations
- Roulette is tractable to analyze, but most things other than human-designed games of chance are not
- Averages are only part of the story
- Important to understand how possible results are distributed
- Regression to the mean often the right explanation
- Coming up
  - More on variance
  - When should we trust a simulation
  - Other examples of Monte Carlo simulation